

LE XUAN HUNG, PhD

Senior Software Engineer ▪ Structural Analysis

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U.S. status: Green Card expected in 2026; no employer visa sponsorship required.

Professional Summary

- Structural engineer and hands-on software developer with more than 10 years in structural engineering and engineering software work since 2018. I turn analysis methods and practical design checks into tools that engineers can use.
- Built desktop and web applications for frame and truss analysis, reinforced-concrete checks, finite element modelling, computational geometry, and construction workflows. My bridge work provides the engineering basis; I am interested in applying it to residential framing and building systems.
- Work in C#, .NET, Python, and JavaScript/React. Use AI-assisted development to prototype faster while reviewing the engineering logic, numerical results, and behaviour of the software I deliver.

Core Skills

Software: C#, .NET, Python, TypeScript, JavaScript, React, Node.js, VBA, AutoLISP, AutoCAD .NET; Git, AWS, SQL, and MongoDB.

Algorithms: Frame and truss analysis; finite element modelling with beam, shell, and plate elements; structural calculations, design checks, and computational geometry.

Structural: Structural mechanics, reinforced concrete and steel, load paths, staged construction, and code-based design verification.

Delivery: Translate engineer and field-user needs into working tools; cross-check numerical outputs, document assumptions, iterate on workflows, and use AI-assisted coding with engineering review.

Selected Engineering Software

For project demos and more detail, please see my GitHub and portfolio links above.

- **PUS Pro Program**
Structural analysis and design software for steel-concrete composite box-girder bridges, with an in-house FEM solver and design checks.
- **Truss Designer**
Analysis and design of steel and timber trusses to US, Korean, and Australian standards.
- **Reinforced Concrete Column Designer**
Biaxial compression-bending design and capacity checks for reinforced-concrete columns under multiple design standards.
- **Steel Bridge Designer**
Web tool under development for steel I-girder and box-girder bridge design.
- **Bridge Alignment**
Horizontal and vertical alignment design for bridges, supporting geometry calculations and site coordination.
- **CastControl**
Web application for segment casting and match-casting geometry. Connects field survey data to design review and automated tolerance checks.

Professional Experience

ENVICO Co., Ltd.

Deputy Project Manager & Engineering Software Developer

Seoul, Korea

Sep 2024 – Present

Key responsibilities

- Lead the company's "vibe coding" program and train staff to use AI-assisted coding to automate structural-design tasks.
- Lead R&D of an integrated bridge-design software workflow, linking geometric modelling, structural analysis, code-based design checks, drawing generation, calculation sheets and technical reports (Excel/PDF), quantity take-offs, and project cost estimates. Use AI to support automation where appropriate, while keeping engineering calculations traceable and verifiable.
- Collaborate with Korean and Swiss research institutes on an AI-assisted digital twin for steel bridges, linking AI-based corrosion indicators with nonlinear structural models for residual-capacity and lifecycle assessment.

Woosung Construction Co., Ltd.

Senior Structural Engineer & .NET Developer

Seoul, Korea

Sep 2018 – Sep 2024

Key responsibilities

- Developed the PUS Pro .NET application for structural modelling, finite element analysis, design verification, quantities, and technical reports.
- Translated structural behaviour, design-code checks, and geometry requirements into calculation workflows; checked software results against engineering analyses.
- Analysed steel and concrete structures, connections, and temporary works, providing practical cases for software development and verification.

Hung Nghiep Construction Consulting Co., Ltd.

Structural Engineer

HCMC, Vietnam

Aug 2007 – Aug 2010

Key responsibilities

- Performed structural and staged-construction analysis and automated repetitive calculations and checks using Mathcad, VBA, AutoLISP, and VB.NET.

Engineering-to-Software Approach

I started programming at university because I wanted structural calculations to be faster and more accurate. Since then, I have used VB, VBA, Mathcad, AutoLISP, MATLAB, Python, C#/.NET, AutoCAD .NET, and TypeScript for engineering work. Structural engineering and software development support each other. To build engineering software, I need to understand the structural principles and calculation method. I also need to know how to choose a suitable software design and implement it efficiently. I enjoy working in both fields and keep learning in both. That is how I became a structural engineer who also develops engineering software.

Education

2012–2018 PhD Civil Engineering, Kyung Hee University

Research on bridge deflection and vibration reliability; peer-reviewed publications.

2010–2012 MEng, Kyung Hee University

2002–2007 BSc Civil Engineering, University of Communications and Transport, Vietnam